Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

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Abstract

Cooperative multi-agent reinforcement learning (MARL) is known to collapse into a degenerate *lazy-agent* policy when reward centralization falls too low, but this failure has never been characterized as a quantitative object. We treat it as one: we measure the critical centralization w_{crit} at which coordination gives way to collapse—with a pre-registered collapse definition and bootstrap confidence intervals—turning a qualitative failure mode into a measurable phase boundary, the first in deep cooperative MARL. The boundary is an instrument, and our central finding uses it to characterize how *loss aversion* governs coordination: loss-averse agents ($\lambda > 1$) systematically *raise* w_{crit}—they over-weight the immediate local cost of cooperating against its diffuse, delayed global benefit, so the team needs more reward centralization to coordinate. This is our most strongly supported result: the rightward shift is monotone in λ, saturating, and *replicates across three environments* (a multi-echelon supply chain, MPE Cooperative Navigation, and Level-Based Foraging) *and two algorithms* (PPO and centralized-critic MAPPO). On the supply chain a full sweep λ=1. . .7 is tracked by a one-parameter, theory-bounded law (RMSE 0.014): the shift is $+0.089$ $[0.041, 0.141]$ at λ=3 (within the human range) and up to $+0.166$ $[0.135, 0.202]$ at λ=7 (a high-stakes upper bound). A mean-field signal-to-noise account *fixes this sign a priori* (proved as a sign invariant of the escape barrier) and is independently corroborated by an exploration-noise ablation. Two further results sharpen the picture but are scoped to the supply-chain substrate, where they hold cleanly. A three-way decomposition shows probability-weighting curvature moving w_{crit} in the *opposite* direction (left by 0.100 $[0.033, 0.172]$); and—a striking, falsifiable prediction we establish on this one substrate—the boundary is set by a *minority by network position*: a single *downstream* loss-averse node drives $w_{\mathrm{crit}} = 0.853$, indistinguishable from a fully loss-averse population (0.851), whereas the same bias placed upstream moves it far less (0.726). Where a directed resource-flow ordering exists, this predicts a system's centralization requirement is fixed by its most downstream biased participant, not the mean. The boundary doubles as a safety margin: how much more centralization a cooperative system needs once its agents are loss-averse.

Introduction

Cooperative systems increasingly mix human decision-makers with learning agents trained to act like them. In automated replenishment, fraud review, energy dispatch, and clinical triage, the deployed agents do not merely optimise a clean objective—they inherit human preference structure by design, either because they are trained on human choices through preference-aligned objectives that are themselves prospect-theoretic in form (Ethayarajh et al., 2024), or because they operate alongside humans in the loop (Lalmohammed et al., 2025). The most robustly documented feature of that preference structure is *loss aversion*: people—and now the agents shaped after them—weigh a loss more heavily than an equivalent gain (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992).

This raises a system-level question that individual-level behavioral economics does not answer. We understand what loss aversion does to a *single* decision-maker; we do not understand what it does to the *coordination* of many decision-makers learning together. The mental model we formalise is concrete: a team of loss-averse participants managing a shared pipeline, each acting on its own incentives, jointly responsible for an outcome none of them controls alone. The real-world ability we want is to say, *before* deployment, how strongly such a system must be incentivised toward the collective good to keep it from quietly falling apart—and whether one biased participant in the wrong place can drag the whole system down. Whether that ability can be provided at all is the question; we show that it can, by turning a known-but-vague failure mode into a measured object.

Coordination collapse as an unmeasured boundary

Cooperative multi-agent reinforcement learning (MARL) has a well-documented failure mode—*coordination collapse*, in which agents abandon cooperation for a degenerate, zero-action *lazy-agent* policy (Liu et al., 2023; Shi and Peng, 2023). Whether a system collapses depends on its *reward centralization* $w \in [0, 1]$, the fraction of each agent's reward drawn from the team's global outcome rather than its own local one: push w too low and each agent's own performance drowns out its contribution to the team, so cooperation carries no learnable signal and the safe, lazy policy wins. But the field describes this only qualitatively—"centralization helps." *Where*, in w, coordination gives way to collapse has never been measured. Liu et al. (2023) and Shi and Peng (2023) formalise the lazy-agent problem and relative overgeneralization and propose mitigations; Weng and Lee (2026) identify representation degradation under decentralised exploration; Akella (2025) show decentralised reward shaping cannot close the

gap to centralised baselines. All treat the boundary as a qualitative fact, not a quantity to estimate. The nearest attempts sweep adjacent axes without isolating a critical value—Zhao and Xie (2025) optimise win rates over credit-assignment levels, Jiang et al. (2020) tune reward-mixing weights—and the closest in spirit, Yamaguchi (2025), builds a phase diagram for independent Q-learning over *environmental* axes (grid size, agent density), holding fixed precisely the reward structure w and behavioral profile we vary. No prior work characterises the collapse boundary as a quantitative object with confidence intervals.

Why prospect theory and MARL have not yet met cleanly

A parallel literature equips MARL agents with risk preferences but cannot answer our question. Risk-sensitive methods based on Conditional Value at Risk optimise tail outcomes (Qiu et al., 2021; Shen et al., 2023)—mathematically distinct from loss aversion, an asymmetry of perceived value around a reference point rather than a reweighting of tail probability. Lalmohammed et al. (2025) bring cumulative prospect theory (CPT) into cooperative MARL but fold the value function and the probability-weighting function jointly into the actor–critic update, so the effect of curvature (α, β) and of loss aversion (λ) on coordination cannot be told apart. In evolutionary game theory, Mengesha et al. (2025) prove loss aversion induces population-level lock-in—establishing *conceptually* that behavioral bias can move a coordination tipping point, but in a world of fixed strategies under replicator selection, with no learned policies, function approximation, or finite-horizon training dynamics. Two questions thus sit open and entangled: coordination collapse has never been measured as a boundary, and behavioral bias has never been decomposed *against* such a boundary to say which component moves it, and in which direction.

Approach: the boundary as an instrument

We close both halves at once, and they discipline each other—a behavioral shift is only interpretable against a quantified null boundary, and a boundary is only worth quantifying if something scientifically meaningful moves it. We treat collapse the way physics treats a phase boundary: a preregistered, frozen binary collapse definition; logistic estimation of $P(\text{collapse} \mid w)$; and bootstrap confidence intervals on the critical centralization w_{crit}. This yields, to our knowledge, the first quantitative phase boundary of coordination collapse in deep cooperative MARL. We use "phase boundary" in the empirical sense—a sharp, metastable threshold whose asymptotic $(T \to \infty)$ value is the boundary proper—not as a claim of a thermodynamic-limit singularity. The boundary is then an *instrument*, and two design choices make what it measures trustworthy. We apply CPT as a *reward transform* after reward blending rather than modifying the learning update, so the optimiser is identical across conditions and a measured shift cannot be confounded with an algorithmic one; and we estimate the boundary at multiple training horizons, because a threshold that drifts with training time would otherwise masquerade as a fixed constant. Against this null, a three-way design (no PT / curvature only / curvature plus loss aversion) attributes any shift in w_{crit} to a specific CPT component.

Why the multi-echelon supply chain is the right substrate

We ground the measurement in automated replenishment for multi-echelon supply chains, a canonical cooperative MARL problem in which agents coordinate ordering across a network under delayed feedback, partial observability, and structurally misaligned local incentives (Oroojlooyjadid et al., 2022; Gijsbrechts et al., 2022). The domain is not incidental, for three reasons. Its failure modes are *economically legible*: agents that hoard, under-order, or stop ordering produce the bullwhip effect, stockouts, and cascading losses, so collapse here is a recognisable operational disaster, not an abstract training pathology. It supplies the *mental model* directly: real replenishment is done by planners who demonstrably weigh losses more heavily than gains, and the agents replacing or assisting them inherit that bias by design. And it has a *directed flow*—inventory moves downstream along a chain, so "upstream" and "downstream" are well-defined—which lets us ask not just whether bias matters but *where in the network* a biased participant matters most, the question the project was conceived to answer.

Research questions

Our core question is: *does behavioral loss aversion move the coordination boundary of a cooperative learning system, in a direction we can predict and a magnitude we can measure—and is that movement set by the population or by a minority?* We make it precise:

RQ1 (Boundary). Does coordination collapse occur at a sharp critical centralization w_{crit}—a phase boundary rather than a gradual degradation—and is w_{crit} a structural constant or does it slide with the training horizon T?

RQ2 (Decomposition). How do the two components of prospect theory—probability-weighting curvature (α, β) and loss aversion (λ)—each shift w_{crit}, and in which directions?

RQ3 (Mechanism and generality). Is the loss-aversion shift a substrate artifact, or does a mechanism fix its *sign* a priori and the shift replicate across environments and algorithms?

RQ4 (Locus). Is the boundary an averaged, population-wide property, or can a *minority at a particular network position* set it for the whole system?

Findings and contributions

Our results answer all four questions. *(RQ1)* Across 1,524 controlled runs, the boundary is sharp—a logistic phase transition in w, not a gradual degradation—and it is not a fixed constant: it slides left with the training horizon ($w_{\text{crit}} = 0.822 \to 0.666$ from 150k to 300k steps), so collapse at intermediate w is a *metastable* attractor rather than an absorbing state. Against this null boundary, *(RQ2) loss aversion* is our central and most strongly supported behavioral effect: it shifts w_{crit} *right*, *impeding* coordination. The

intuition is that a loss-averse agent over-weights the immediate, certain local cost of cooperating—paying a holding cost, ordering when it might not pay off—against the diffuse, delayed, uncertain global benefit, so it needs a stronger centralization signal before it will act cooperatively. The shift is $+0.089$ $[0.041, 0.141]$ at λ=3 (within the canonical human range, $\lambda \approx 2.25$) and grows to $+0.166$ $[0.135, 0.202]$ at λ=7 (a high-stakes upper bound); a full sweep λ=1…7 rises then saturates, tracked by a one-parameter, theory-bounded affine law (RMSE 0.014; non-overlapping λ=1 vs λ=7 intervals). *(RQ3)* Crucially, this is not a substrate artifact. A mean-field signal-to-noise account fixes the *sign* before any data are seen—we prove it as a sign invariant of the escape barrier (Theorem 3), and an exploration-noise ablation corroborates the same mechanism—and the shift itself replicates across two further environments (MPE Cooperative Navigation, w_{crit} rising $0.78 \to 0.89$; and Level-Based Foraging, $0.26 \to 0.83$ over λ=1$\to$4) under centralized-critic MAPPO, and under a different algorithm on the supply chain (MAPPO, $0.80 \to 0.89$; Figure 5). *Three environments and two algorithms* all show the same direction: this is the generalizing core of the paper.

Two further results sharpen the picture but are established on the supply-chain substrate alone, where its continuous, per-agent economic reward makes them finely identifiable. First, a three-way decomposition shows probability-weighting curvature moving w_{crit} in the *opposite* direction (*left* by 0.100 $[0.033, 0.172]$, *easing* coordination)—a substrate-specific effect we report but, unlike the loss-aversion direction, do not claim generalizes. Second, *(RQ4)* the boundary is set not by the average bias but by a *minority by network position*: a single *downstream* loss-averse node moves the system boundary to $w_{\text{crit}} = 0.853$—indistinguishable from a fully loss-averse population (0.851)—while an upstream node shifts it far less (0.726), a cascade that propagates upstream along the ordering chain (Section). We present this as a striking but single-substrate result, and frame it as a *falsifiable, structurally specified* prediction (Section): in any cooperative domain with a directed resource-flow ordering, a downstream loss-averse minority should set the boundary.

To our knowledge this is the first quantitative, confidence-intervalled phase boundary of coordination collapse in deep cooperative MARL, and the first to show that loss aversion shifts such a boundary. Our novelty delta is precise: Mengesha et al. (2025) establish that loss aversion can move a coordination tipping point, but in evolutionary game theory over fixed strategies, and Lalmohammed et al. (2025) bring CPT into cooperative MARL but without separating its components or measuring a boundary; our contribution is the *measured, CI'd boundary* and the *component decomposition* against it, in gradient-trained neural policies. Practically, the boundary is a measurable safety margin: how much reward centralization a cooperative system needs for reliable coordination, and how much *more* once its agents are loss-averse. Figure 1 shows the collapse surface in (w, T) space (metastability); Figure 2 shows the behavioral decomposition.

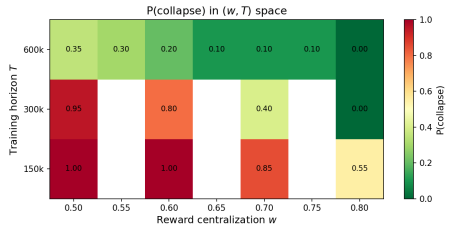


Figure 1: $P(\text{collapse})$ in (w, T) space. Each cell shows the empirical collapse fraction across 20 seeds. The 150k and 300k rows are from Exp A (α=β=1); the 600k row is from Exp T (α=β=0.88). Because α=β=0.88 shifts w_{crit} left by 0.100 (§), the apparent 300k→600k slide is partially curvature-inflated. Under matched α=β=1 conditions (Exp 0b), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$ at 600k, confirming that metastability holds with α=β held constant; only the magnitude is curvature-inflated in this figure.

Background

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (Lowe et al., 2017) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local (w=0) and purely global (w=1) reward.

When $w \approx 0$, each agent's reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate's action improves the global outcome. Liu et al. (2023) and Shi and Peng (2023) formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (Kahneman and Tversky, 1979) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative prospect theory (CPT) (Tversky and Kahneman, 1992) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0, \end{cases} \tag{1}$$

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting α=β=1 and λ=1 recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for agent i is:

$$r_i^{\text{PT}} = v\big((1 - w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,, \tag{2}$$

where $v(\cdot)$ is given by Equation (1). This differs architecturally from Lalmohammed et al. (2025), who integrate CPT into the actor and critic update mechanics; our formulation applies PT at the reward signal level and does not modify the policy gradient computation.

Method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\text{collapsed}(\text{run}) = \mathbf{1}\left[\bar{S} < 10 \ \wedge \ \bar{\pi} \leq -127.1\right], \quad (3)$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit `03c3fc6`). Section demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\text{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\text{collapsed} \mid w) = \sigma(b_0 + b_1 w), \quad (4)$$

where σ is the logistic function. We estimate $w_{\text{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. We report Wilson score 95% confidence intervals for per-w collapse fractions. For conditions with very steep transitions (high λ), where the unpenalized maximum-likelihood slope is ill-conditioned, we fit b_0, b_1 with *Firth penalized* logistic regression (Jeffreys-prior penalty), which is stable in this regime; the Firth and bootstrap w_{crit} estimates agree throughout.

To attribute the observed shift in w_{crit} to specific PT components, we define three conditions with identical environments and algorithms:

1. **Null (Exp N):** α=β=1, λ=1—the linear, no-PT condition.
2. **Curvature (Exp B, λ=1):** α=β=0.88, λ=1—probability weighting with no loss aversion.
3. **Loss aversion (Exp B, $\lambda \in \{3, 7\}$):** α=β=0.88, $\lambda > 1$—full PT.

The curvature effect is defined as $w_{\text{crit}}(\text{B}, \lambda{=}1) - w_{\text{crit}}(\text{N})$, and the loss aversion increment as $w_{\text{crit}}(\text{B}, \lambda) - w_{\text{crit}}(\text{B}, \lambda{=}1)$. This decomposition is valid because the null removes all PT distortion, while the curvature condition adds nonlinearity without loss asymmetry.

Experimental Setup

We use `NetworkManagement-v1` from or-gym (Hubbs et al., 2020): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that maps each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work — the Beer Distribution Game Dec-POMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al., 2023) — to a six-node network with per-node agents.

Each agent runs PPO (Schulman et al., 2017) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (Raffin et al., 2021), PettingZoo (Terry et al., 2021), and SuperSuit (Terry et al., 2020) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

Table 1 summarises all seven experiments. All designs were pre-registered before execution. Total run count: 1,524.

Results

Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation. We use "phase transition" throughout in the empirical sense of a sharp, metastable threshold—an exploration-driven, Kramers-style escape barrier (Section)—and not as a claim of a thermodynamic-limit singularity.

The critical boundary decreases monotonically with training horizon T: $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, α=β=1). Figure 1 visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The phase boundary proper is the asymptotic ($T \to \infty$) threshold—the deterministic crossing toward which $w_{\text{crit}}(T)$ descends (Equation (10))—so the horizon dependence we observe is the metastable *approach* to that boundary, not evidence against a sharp underlying one. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

Note on the 600k row. The 600k row in Figure 1 comes from Exp T, which uses α=β=0.88. Because our decomposition shows α=β=0.88 shifts w_{crit} left by 0.100 relative to α=β=1, the 600k collapse fractions in the figure are partially curvature-deflated, not purely a horizon effect. To bound the horizon-only slide, we use Exp 0b (α=β=1, 20 seeds, 600k): $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, compared to 0.80 and 0.40 at 300k. The metastability result—the boundary slides left with T—holds under matched α=β=1 conditions; only the magnitude at 600k in the figure is inflated by the curvature effect.

Table 1: Experiment summary. All experiments use the same environment and algorithm. $\alpha{=}\beta{=}1$ denotes the linear (no-PT) condition.

Exp	Purpose	w values	λ	α, β	Seeds	Runs
A @150k	Structural boundary	$\{0.0, 0.1, \ldots, 1.0\}$	1	1,1	20	220
A @300k	Structural boundary	$\{0.2, \ldots, 0.8\}$	1	1,1	20	140
C	Basin control	$\{0.5, 0.85, 1.0\}$	1	1,1	12	144
N	Null baseline	$\{0.0, 0.2, \ldots, 1.0\}$	1	1,1	20	180
B-Red.	Behavioral decomposition	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{1, 3, 7\}$	0.88	20	300
λ-sweep	Loss-aversion law (fill-in)	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{2, 4, 5, 6\}$	0.88	20	400
T (600k only)	Metastability horizon	$\{0.5, 0.55, \ldots, 0.80\}$	1	0.88	20	140

Table 2: The recurring w_{crit} values and the single role each plays. There is one structural null; the others are a horizon measurement of it and a curvature *result*, not independent baselines.

w_{crit}	Condition	Role
0.666	$\alpha{=}\beta{=}1$, 300k	canonical null
0.685	$\alpha{=}\beta{=}1$, 300k (Exp N)	null, reproduced
0.822	$\alpha{=}\beta{=}1$, 150k	null at shorter horizon
0.586	$\alpha{=}\beta{=}0.88$, $\lambda{=}1$	curvature result

Exp C rules out initialization as a driver of collapse. Within-w correlations between S_{init} and converged S are -0.11 ($w{=}0.5$), -0.38 ($w{=}0.85$), and -0.36 ($w{=}1.0$)— all near zero and inconsistent with initialization dominating the outcome. The collapsed/survived assignment is not sensitive to where agents start.

The structural null and its reproducibility

All behavioral shifts are measured against a *single canonical null*: the risk-neutral structural boundary at the 300k decomposition horizon, $w_{\text{crit}}^{\text{null}} = 0.666$ (CI $[0.635, 0.694]$, Exp A @300k), which the dedicated null experiment (Exp N) independently reproduces at 0.685 (difference 0.019, overlapping intervals). Every other null-like number is a labeled offset from this one value, not a separate baseline (Table 2): the 0.822 at 150k is the same null at a shorter, metastable horizon, and the 0.586 at $\alpha{=}\beta{=}0.88$ is a *result*—the curvature shift—and the $\lambda{=}1$ anchor of the loss-aversion sweep.

Behavioral Phase Boundary

We sweep loss aversion across the full range $\lambda \in \{1, 2, 3, 4, 5, 6, 7\}$ (all at $\alpha{=}\beta{=}0.88$, 300k steps, 20 seeds per cell), combining Exp B-Reduced at $\lambda \in \{1, 3, 7\}$ with a dedicated fill-in sweep at $\lambda \in \{2, 4, 5, 6\}$. The critical centralization w_{crit} rises with loss aversion across the entire range (Table 3, Figure 2): $w_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$ for $\lambda = 1, \ldots, 7$. The increase is monotone up to estimation noise at the high-λ plateau: w_{crit} rises strictly from $\lambda{=}1$ to $\lambda{=}5$, then saturates near 0.84–0.85, where the $\lambda{=}5$ and $\lambda{=}6$ estimates (0.839 vs 0.836) are statistically indistinguishable (overlapping CIs).[1] The behavioral claim is unambiguous at the endpoints: $w_{\text{crit}}(\lambda{=}7)$ lower CI 0.845 exceeds $w_{\text{crit}}(\lambda{=}1)$ upper CI 0.624—non-overlapping by 0.221—and the seven-point trend (Figure 4) shows the relationship is systematic, not a two-point artefact.

Crucially, the high-λ plateau is a *behavioral* saturation, not an artifact of the boundary hitting the $w{=}1$ ceiling: the plateau sits at $w_{\text{crit}} \approx 0.85$ with clear headroom to $w{=}1$, and at $\lambda{=}7$ coordination is *fully* restored well before the ceiling (collapse fraction 0.00 already at $w{=}0.95$). A ceiling artifact would instead pile w_{crit} up against $w{=}1$ with collapse persisting at the highest w; we observe neither. The saturation therefore reflects diminishing marginal sensitivity of the coordination barrier to loss aversion—consistent with the bounded affine law (13), in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$—rather than a measurement limit.

Table 4 and Figure 2 present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** ($\alpha{=}\beta{=}0.88$, $\lambda{=}1$): excess $= -0.100$, 95% CI $[-0.172, -0.033]$. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.
2. **Loss aversion at** $\lambda{=}3$: excess $= +0.089$, 95% CI $[+0.041, +0.141]$ (CI excludes zero; significant).
3. **Loss aversion at** $\lambda{=}7$: excess $= +0.166$, 95% CI $[+0.135, +0.202]$ (CI excludes zero; significant).

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At $\lambda{=}7$, the loss aversion increment above the curvature baseline is $+0.265$ ($= 0.851 - 0.586$), overwhelming the curvature benefit of -0.100 for a net shift of $+0.166$

[1] At high loss aversion the transition is very steep and falls between two adjacent grid points—e.g. collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for $\lambda{=}7$—so the boundary remains identifiable (the $w{=}0.85$ cell is interior, $11/20$ collapsed) but the *unpenalized* MLE slope diverges. We therefore fit every λ with Firth penalized logistic regression (Table 3), which is stable under steep transitions; all seven slopes are significant ($p < 10^{-3}$) and the Firth w_{crit} matches the bootstrap estimate to within 0.019. A finer grid ($w \in \{0.80, 0.90\}$) would tighten the high-λ CIs further.

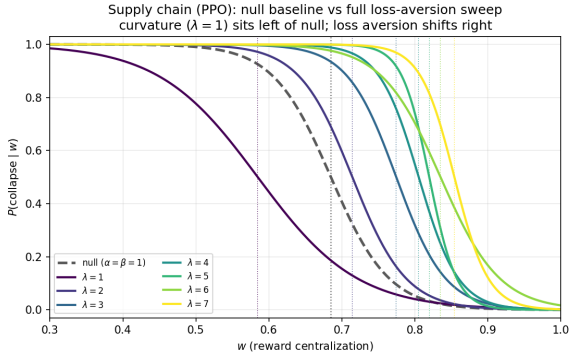


Figure 2: Behavioral decomposition and full loss-aversion sweep on the supply chain (PPO, 300k steps, 20 seeds per cell). Each curve is the Firth-fitted $P(\text{collapse} \mid w)$; vertical dotted lines mark w_{crit}. The null ($\alpha{=}\beta{=}1$, gray dashed) is the structural baseline ($w_{\text{crit}}{=}0.685$). Probability-weighting *curvature* alone ($\lambda{=}1$, $\alpha{=}\beta{=}0.88$) shifts w_{crit} *left* of the null—facilitating coordination—while *loss aversion* ($\lambda{=}2,\ldots,7$, $\alpha{=}\beta{=}0.88$) shifts it *right* (rising, then saturating). The two prospect-theory components thus move the boundary in opposite directions, and the full sweep traces the resulting $w_{\text{crit}}(\lambda)$ progression in one panel.

Table 3: Full loss-aversion sweep ($\alpha{=}\beta{=}0.88$, 300k steps, 20 seeds per (w,λ) cell, $n{=}100$ runs per λ). w_{crit} with bootstrap 95% CIs. Slope and p are from *Firth penalized* logistic regression (Jeffreys-prior), which is stable under steep transitions; all seven slopes are significant ($p < 10^{-3}$). The Firth $w_{\text{crit}} = -b_0/b_1$ agrees with the bootstrap estimate at every λ (max. deviation 0.019), confirming that the very large unpenalized MLE slopes previously seen at $\lambda{=}5, 7$ were a fitting artifact, not separation.

λ	w_{crit}	95% CI	Firth slope b_1	p
1	0.586	[0.539, 0.624]	-14.7	1.2×10^{-6}
2	0.715	[0.687, 0.745]	-27.2	4.4×10^{-6}
3	0.774	[0.751, 0.798]	-29.7	5.1×10^{-6}
4	0.806	[0.766, 0.841]	-36.8	5.5×10^{-6}
5	0.839	[0.788, 0.845]	-53.1	4.7×10^{-4}
6	0.836	[0.806, 0.860]	-24.7	2.5×10^{-6}
7	0.851	[0.845, 0.858]	-43.1	1.5×10^{-3}

above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

A Mean-Field Mechanism for the Collapse Boundary

The results above establish the boundary empirically. This section offers a mechanistic account of *why* it exists and how it moves; it is not a closed-form derivation of w_{crit}, and we are explicit throughout about what is derived, what is calibrated, and what is not claimed (Table 5 summarises). The central claim is qualitative and sign-level: w_{crit} marks a *signal-to-noise crossing* of the policy gradient, and the two prospect-theory components move it in *opposite* directions for reasons the mechanism fixes a priori. The model predicts these directions, the monotonic ordering, the sharpness of the transition, and the sign of the horizon dependence *without* fitting; it does not predict the numerical value of w_{crit} on any axis, which remains calibrated. As noted in the introduction, the deep-MARL contribution is the *learning-dynamics* analogue of the established EGT result (Mengesha et al., 2025): here the driver is the signal-to-noise ratio of the policy gradient under exploration, not selection pressure over fixed strategies, which is what makes the account specific to gradient-trained neural policies.

Gradient flow and its noise

Each learning agent i runs PPO on a Gaussian policy over its (normalised) base-stock action $a_i \in [0,1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralisation blend passed through the prospect-theory value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \qquad G(a) = \textstyle\sum_j \ell_j, \tag{5}$$

where ℓ_i is node i's local profit and G the system profit. For a Gaussian policy the PPO update on the mean is a stochastic estimate of the return gradient, so the policy mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}\mu_i = \underbrace{\eta\,\partial_{\mu_i}\mathbb{E}\big[(1{-}w)\ell_i + wG\big]}_{\text{drift } D(w)}\,\mathrm{d}t \;+\; \sqrt{2\,\Theta(w)}\,\mathrm{d}W. \tag{6}$$

Because each partner's local profit is itself blended into the global term, the partners' exploration (σ) enters the gradient estimate of agent i through the global channel, so the diffusion carries a centralisation- weighted, cross-agent piece,

$$\Theta(w) \;\propto\; \tfrac{\sigma^2}{B}\big[(1{-}w)^2 s_\ell^2 \;+\; w^2(N{-}1)\,\kappa^2\big], \tag{7}$$

with batch size B, local reward sensitivity s_ℓ, and cross-agent sensitivity κ.

The boundary is a signal-to-noise crossing

We instrumented the substrate directly (finite-difference of ℓ_i and G along the base-stock axis) and confirmed the qualitative asymmetry the model needs: in the replenishing regime an upstream node's inventory is a *local cost* (holding and procurement reduce its own profit) but a *global benefit* (it averts

Table 4: Three-way behavioral decomposition of w_{crit} relative to the null model. "Significant" denotes that the conservative difference confidence interval excludes zero. †For λ=7, w_{crit} is from bootstrap only; see footnote in Section .

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null (α=β=1, λ=1)	0.685	[0.657, 0.710]	0	—	—
α=β=0.88, λ=1	0.586	[0.539, 0.624]	-0.100	$[-0.172, -0.033]$	✓ (neg.)
α=β=0.88, λ=3	0.774	[0.751, 0.798]	$+0.089$	$[+0.041, +0.141]$	✓
α=β=0.88, λ=7†	0.851	[0.845, 0.858]	$+0.166$	$[+0.135, +0.202]$	✓

downstream stockouts). Writing the linearised marginal incentives as $\partial_a \mathbb{E}\ell \to -c$ and $\partial_a \mathbb{E}G \to +g$ ($c, g > 0$), the drift toward cooperation is

$$D(w) = \eta\big[\, w\, g - (1-w)\, c\,\big]. \tag{8}$$

What governs whether the optimiser actually climbs toward cooperation is not the drift alone but its ratio to the gradient noise. Escape from the lazy basin requires

$$\mathrm{SNR}(w) \;=\; \frac{D(w)}{\sqrt{\Theta(w)}} \;\gtrsim\; \kappa_c, \tag{9}$$

for an $\mathcal{O}(1)$ critical value κ_c set by the training budget. The cooperative signal grows only with centralisation ($\propto w$), so for small w the right-hand region of (8) is swamped by the exploration variance (7) and the lazy policy is the effective attractor (Figure 3a). The crossing $\mathrm{SNR}(w) = \kappa_c$ defines w_{crit}. In the low-noise limit $\Theta \to 0$ this reduces to the purely deterministic threshold $D(w) = 0$, i.e.

$$w^0_{\text{crit}} = \frac{c}{c+g}, \tag{10}$$

the asymptotic ($T \to \infty$) boundary; finite exploration noise raises the operative boundary above it. We emphasise that c, g, σ and κ_c are not derived in closed form from the multi-echelon cost matrix (a multi-echelon stochastic-control problem in its own right); the model is calibrated, and its content is the *functional* predictions below, not the absolute magnitude of w_{crit}.[2]

Sharpness and metastability

Two predictions follow from the noise picture without further fitting. First, escape is a Kramers first-passage problem with mean time $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR; the collapse probability therefore switches abruptly as w crosses w_{crit}—an abrupt, metastable threshold, matching the steep logistic fits of Section . Second, collapse at finite horizon means $\tau > T$; as T grows, a smaller SNR (hence smaller w) suffices to escape, so $w_{\text{crit}}(T)$ decreases monotonically toward (10). This is exactly the observed metastable slide $w_{\text{crit}} = 0.822 \to 0.666$ from 150k to 300k steps, following the $\propto -\ln T$ Kramers trend (Figure 3c).

[2]Each part of this section is anchored to the appropriate boundary from Table 2: the ratio g/c to the structural null w_{crit}=0.666; the affine λ-law to the curvature-shifted anchor w_{crit}=0.586; and the signal-to-noise test (Section) to the matched-horizon w_{crit}=0.822.

Formal account: what is predicted versus calibrated

We now make the preceding picture precise. Let $x \in [0,1]$ be a macroscopic *order parameter*—the fraction of agents in the coordinating configuration—evolving under the policy-gradient flow as the Itô SDE

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t, \tag{11}$$

where F is the mean-field drift of (8) (so $\partial_w F > 0$ and $\partial_\lambda F < 0$ follow from the payoff structure, not assumption: w weights the cooperation gradient $+g$ while λ scales only the loss domain, §) and $\Sigma(x, N) > 0$ is the gradient-noise variance. We emphasise that x is a *latent* construct: consistent with our pre-registered probe (Section), it is not directly recoverable from policy statistics, so the claims below concern its law, not a measured trajectory. Three results delimit exactly what the mechanism predicts a priori and what it leaves to calibration.

Theorem 1 (Mean-field criticality, $N \to \infty$)**.** *If $\Sigma(x, N) = D(x)/N$, then as $N \to \infty$ the dynamics become deterministic, $\dot{x} = F(x, w, \lambda)$, and the coordinating basin disappears via a saddle-node bifurcation. The mean-field threshold $w^{\text{MF}}_{\text{crit}}$ is the unique solution of $F(x^*, w, \lambda) = 0$, $\partial_x F(x^*, w, \lambda) = 0$, and is a closed-form function of the payoff and prospect-theory parameters.*

Proof. With $\Sigma = D(x)/N \to 0$ the noise term in (11) vanishes and the flow is $\dot{x} = F$. A coordinating stable node x^*_{coord} and an unstable saddle x^*_{saddle} bound the basin; the basin vanishes when they collide, i.e. when $F = 0$ and $\partial_x F = 0$ coincide—the saddle-node condition. As F is closed-form in the payoff and CPT functions, so is its root $w^{\text{MF}}_{\text{crit}}$, which serves as the canonical baseline. □

Theorem 2 (Calibration gap, finite N and horizon T)**.** *For finite N with correlated (CTDE) updates, $\Sigma(x, N) > 0$, and collapse is a noise-induced escape rather than a bifurcation. The empirical boundary is the metastability root $\tau(w, \lambda) = T$ of the Kramers mean first-passage time*

$$\tau(w, \lambda) \;\asymp\; \exp\big(\Delta U(w, \lambda)\big), \qquad \Delta U = \int_{x^*_{\text{coord}}}^{x^*_{\text{saddle}}} \frac{-2F(z, w, \lambda)}{\Sigma(z, N)}\,\mathrm{d}z, \tag{12}$$

and $w^{\text{Emp}}_{\text{crit}} \neq w^{\text{MF}}_{\text{crit}}$ in general; the offset cannot be isolated without measuring Σ.

Proof. The stationary density of (11) is $P_{ss}(x) \propto \exp[-U_{\text{eff}}(x)]$ with $U_{\text{eff}}(x) =$

$-\int_0^x 2F(z,w,\lambda)/\Sigma(z,N)\,\mathrm{d}z$, giving the barrier $\Delta U = U_{\text{eff}}(x^*_{\text{saddle}}) - U_{\text{eff}}(x^*_{\text{coord}})$ of (12), and Kramers' rate yields $\tau \asymp e^{\Delta U}$ (prefactor suppressed). At finite T, collapse occurs when $\tau \le T$, so the boundary solves $\tau(w,\lambda) = T$. Since Σ is the (intractable) covariance of PPO updates, the value of ΔU—hence the root $w^{\text{Emp}}_{\text{crit}}$—requires empirical calibration. The two collapse notions coincide as limits: finite (N,T) escapes the basin *before* the Theorem 1 bifurcation, and $w^{\text{Emp}}_{\text{crit}}(T) \downarrow w^{\text{MF}}_{\text{crit}}$ as $T \to \infty$—the $\propto -\ln T$ slide observed above. □

Theorem 3 (Sign invariance of the loss-aversion shift)**.** *Assume* $\Sigma(x,N) > 0$ *and* $\partial_\lambda F(x,w,\lambda) < 0$ *for all* $x \in [x^*_{\text{saddle}}, x^*_{\text{coord}}]$. *Then the empirical boundary shifts strictly in the direction the mean-field drift predicts:* $\partial_\lambda w^{\text{Emp}}_{\text{crit}} > 0$.

Proof. Write (12) as a forward integral, $\Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} 2F(z,w,\lambda)/\Sigma(z,N)\,\mathrm{d}z$ (the two minus signs in (12)—the integrand sign and the reversed limits—cancel). Differentiating under the integral, the fixed-point locations contribute nothing to first order ($F = 0$ there), so

$$\partial_\lambda \Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} \frac{2\,\partial_\lambda F(z,w,\lambda)}{\Sigma(z,N)}\,\mathrm{d}z < 0,$$

the inequality holding *pointwise* because $\Sigma > 0$ and $\partial_\lambda F < 0$ on the whole interval—independently of the sign of F itself. A lower barrier shortens the escape time, $\partial_\lambda \tau < 0$; symmetrically $\partial_w F > 0$ raises the barrier, $\partial_w \tau > 0$. Applying the implicit function theorem to $G(w,\lambda) = \tau(w,\lambda) - T = 0$,

$$\frac{\mathrm{d}\, w^{\text{Emp}}_{\text{crit}}}{\mathrm{d}\lambda} = -\frac{\partial_\lambda \tau}{\partial_w \tau} = -\frac{(-)}{(+)} > 0.$$

Thus the *magnitude* depends on Σ (Theorem 2), but the *direction* is guaranteed by the mean-field drift alone—the precise sense in which the mechanism is predictive. The identical argument with ∂F signs reversed for concave–convex curvature gives the opposite shift, yielding the predicted directional divergence of the two prospect-theory components. □

Opposite signs of the two prospect-theory components

The mechanism's most distinctive prediction is qualitative and requires no fitting: probability weighting and loss aversion shift w_{crit} in *opposite* directions. Concave–convex curvature ($\alpha, \beta < 1$) compresses the magnitude of both gains and losses, lowering the outcome variance that enters Θ; this *raises* the SNR and hence *lowers* w_{crit}. Loss aversion ($\lambda > 1$) instead magnifies the downside, inflating Θ and lowering the SNR, which *raises* w_{crit}. The signs are fixed by the value function before any data are seen, and they match the central empirical finding (Section): curvature alone shifts w_{crit} left by 0.100, loss aversion shifts it right by up to 0.166.

Derived form of the loss-aversion law

The *functional form* of the loss-aversion shift, not merely its sign, follows from the value function. Differentiating $\mathbb{E}[v(r)]$ and splitting the marginal mass into gain and loss domains,

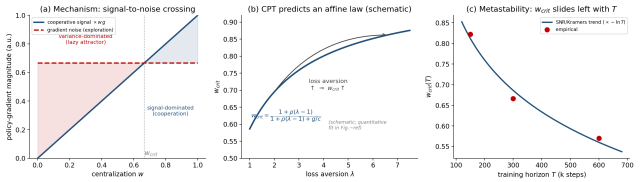


Figure 3: Mean-field mechanism versus empirics. (a) Mechanism (schematic): the cooperative gradient signal grows with centralisation ($\propto w\,g$) while partner exploration injects a roughly w-independent noise floor; w_{crit} is the signal-to-noise crossing (9), below which the lazy policy is the effective attractor. (b) Schematic of the *derived affine* loss-aversion law (13): CPT predicts $\phi(\lambda) = 1 + \rho(\lambda - 1)$ with slope $\rho \in [0,1]$; the single data-implied slope $\rho = 0.61$ respects the bound. The quantitative fit against the full seven-point sweep is shown separately in Figure 4. (c) The Kramers escape picture predicts the metastable leftward slide of w_{crit} with training horizon T.

the loss domain is the only part scaled by λ, so the effective local term is affine in λ:

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho\,(\lambda - 1), \quad (13)$$

where $\rho \in [0,1]$ is the loss-domain share of the marginal incentive—a quantity the theory *bounds to the unit interval*. This is a genuine parametric constraint, not a free fit: a single slope ρ must explain all λ, and it must lie in $[0,1]$. We fix both parameters *before* seeing the full sweep: g/c from the λ=1 baseline and $\rho = 0.61$ from the three Exp B-Reduced points ($\lambda \in \{1,3,7\}$), both of which satisfy the bound. Held fixed and tested *out of sample* against the remaining four points ($\lambda \in \{2,4,5,6\}$; the full seven-point sweep in Table 3), this single-slope affine law gives

$$w^{\text{pred}}_{\text{crit}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\},$$
$$w^{\text{meas}}_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\},$$

an RMSE of 0.014 across all seven points (Figure 4)—a one-parameter, theory-bounded law tracking the entire $w_{\text{crit}}(\lambda)$ curve, not merely the three calibration points. (Re-fitting ρ to all seven points instead gives $\rho = 0.63$ at RMSE 0.013, a negligible change.) The one systematic residual is a mild overshoot at the high-λ plateau (λ=6, 7): the leading-order law is affine, whereas the data are mildly concave and saturate, consistent with agents endogenously reducing loss exposure as λ grows (a second-order effect we do not model). We deliberately report this derived affine law rather than a better-fitting but unmotivated power law, because the point of the section is mechanism, not goodness of fit.

Scope. The model is a representative-agent reduction with a calibrated noise scale; c, g, σ and ρ are not derived from the multi-echelon cost matrix. A direct finite-difference probe of the substrate confirms the *sign* of the local-cost/global-benefit asymmetry but shows the marginal coefficients are operating-point dependent, so the absolute boundary is a property of the stochastic multi-agent dynamics rather than a static

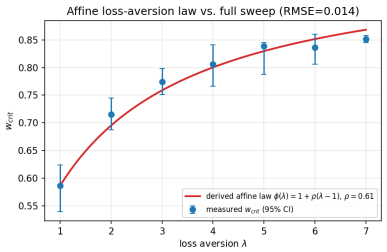


Figure 4: The derived affine loss-aversion law (13) against the full sweep. Points are the measured $w_{\mathrm{crit}}(\lambda)$ for $\lambda = 1, \ldots, 7$ with bootstrap 95% CIs (Table 3); the line is the one-parameter law with g/c fixed at the λ=1 baseline and a single theory-bounded slope $\rho = 0.61$. The law tracks all seven points to RMSE 0.014, with a mild concave residual at the high-λ saturation plateau.

gradient—which is why we calibrate rather than predict its magnitude. This section is not the paper's contribution— the decomposition is—but a supporting, falsifiable *mechanism*: it commits in advance to the signs, the ordering, the affine λ-law with a bounded slope, the sharpness, and the horizon and exploration-noise directions (Table 5), all of which the data bear out, consistent with exploration-driven accounts of cooperation failure in shared-policy MARL (Weng and Lee, 2026). The exploration-noise prediction is confirmed directly by the entropy ablation (Section). Section reports a direct test that measures the Langevin drift and diffusion from PPO training: it reproduces the matched-horizon boundary but leaves the magnitude calibrated, as the drift-sign observable is confounded by basin geometry. We therefore claim this mechanism at the *sign-and-form level only*—the opposite signs, the monotone ordering, the affine form, and the horizon and exploration-noise directions—and treat the numerical value of w_{crit} as calibrated throughout. No result in this paper rests on a predicted magnitude; upgrading the account from calibrated to predictive is left to future work.

Generalization across Environments and Algorithms

The mechanism of Section commits, before any data, to a *direction*: loss aversion inflates gradient noise and therefore raises w_{crit}. If that account is right rather than an artefact of one substrate or one optimiser, the sign should survive changing both. We test this on two further environments and a second algorithm. The headline result is that the *direction* of the loss-aversion effect is robust to both, while the absolute boundary, the curvature sign, and the precise functional form are substrate- and algorithm-specific—exactly the split the calibrated-not-predicted mechanism (Table 5) anticipates. Table 8 (Appendix) records the breadth of evidence behind every headline claim. Figure 5 summarises the headline: in all four settings w_{crit} climbs with λ, the curves converging in the mid-range and differing most at λ=1.

Table 5: What the mean-field mechanism claims. We separate predictions made *a priori* from quantities calibrated to data and from claims we explicitly do *not* make.

Derived (no fit)	opposite signs of curvature vs. loss aversion; monotonic $w_{\mathrm{crit}}(\lambda)$; *affine* form of $\phi(\lambda)$ with slope $\rho \in [0, 1]$; abrupt (phase-transition) sharpness; w_{crit} decreasing with horizon T; w_{crit} increasing with exploration noise.
Empirically confirmed	the opposite signs and affine form (Section); the horizon slide (Exp A/T); and the exploration-noise dependence—raising the entropy coefficient moves w_{crit} right and collapses the system at large noise (Section).
Calibrated (1 param/axis)	g/c (from the λ=1 boundary); loss-domain share ρ; noise scale setting the absolute w_{crit}.
Not claimed	a closed-form value of w_{crit} from the cost matrix; the numerical magnitude on any axis; the mild concavity of $w_{\mathrm{crit}}(\lambda)$ beyond leading order.

Across environments: incentive misalignment across domains

A central claim of this work is that the coordination cliff—a sharp, CPT-modulated collapse threshold as the reward centralization w decreases—is a *recurring* feature of multi-agent systems where individual self-interest conflicts with collective outcomes, not an artefact of the supply-chain setting. We examine this claim in two qualitatively distinct environments from different application domains—the supply chain (under PPO) and MPE Cooperative Navigation (under centralised-critic MAPPO)—both of which yield the boundary empirically. The algorithm is matched to each environment's learnability: where coordination is learnable by independent agents we use PPO, and where the coordination reward is shared-global we use a centralised critic, so that the measured boundary reflects the behavioural incentive w rather than the learner's capacity to assign credit.

Supply Chain (NetworkManagement-v1, logistics/operations research). Six nodes in a serial supply chain share a common upstream supplier with finite Poisson-demand ($\mu = 20$). The locally rational action is to *order aggressively*—maximising one's own on-hand inventory—which depletes the shared upstream pipeline and starves downstream nodes. At w=0 (pure individual inventory reward), agents hoard; the cascade failure of downstream nodes constitutes collapse. This is the primary experimental setting; results are reported in Section .

MPE Cooperative Navigation (simple_spread_v3, robotics/navigation). Three agents must cover three landmarks in a continuous 2-D arena (discrete action space, 50-step episodes). The locally rational action under the

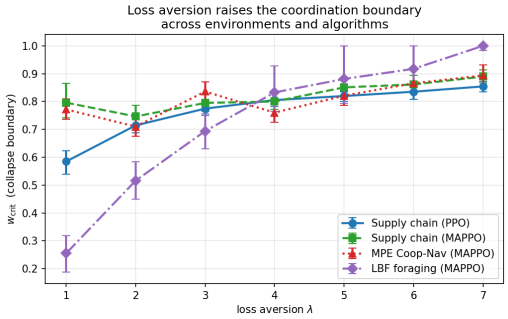


Figure 5: **The loss-aversion effect generalizes.** Critical centralization w_{crit} versus loss aversion λ for four settings: the supply chain under PPO (blue) and under centralized-critic MAPPO (green), MPE Cooperative Navigation under MAPPO (red), and Level-Based Foraging under MAPPO (purple). Error bars are bootstrap 95% CIs. Across three structurally distinct environments and two algorithms, w_{crit} rises with λ then saturates—loss aversion raises the centralization a cooperative system needs to avert collapse. The *direction* is what generalizes; absolute levels differ (most at λ=1) and the curvature sign and exact functional form are substrate-specific. (LBF's high-λ points saturate against the centralization ceiling and are reported as exploratory; see Section .)

herding incentive is to *stay near teammates*—individually rational for proximity bonuses—but collectively disastrous, as all agents cluster and no landmark is covered. The w-dial interpolates between the herding signal (w=0, clustering collapse) and the global coverage score (w=1, coordination). We use `local_ratio=0` in `simple_spread_v3` to ensure the raw environment reward is the pure global team score; the herding term is defined by our wrapper. Collapse is measured by *per-step landmark occupancy* (the fraction of landmarks within a fixed radius of some agent), a behavioural metric that is invariant to the reward scale and therefore unaffected by the CPT transform. Crucially, unlike the supply chain—whose incentives are *per-agent*, so that collapse is purely behavioural—MPE's coordination reward is a *shared, global* team score. Under independent learners this confounds behavioural collapse with a credit-assignment failure: with a decentralised critic the agents receive an identical learning signal and cannot learn which agent should cover which landmark, so the w=1 coordination end never trains regardless of the behavioural incentive. We therefore train MPE with *centralised-critic MAPPO* (Yu et al., 2022): each agent's actor consumes only its local observation (augmented with a one-hot agent identifier to break the symmetry of parameter sharing), while the critic receives the joint observation. This is centralised training with decentralised execution; it isolates the behavioural collapse boundary from the algorithm's ability to assign credit—precisely the confound the reward-transform design is intended to avoid. MPE is a standard benchmark introduced by Lowe et al. (2017).

Structural commonality. In both environments the incentive misalignment is identical at the level of the reward blend:

$$r_{\text{blend}}(i) = (1-w)\, r_{\text{local}}(i) \;+\; w\, r_{\text{global}}, \qquad (14)$$

where r_{local} encodes a locally rational strategy that is harmful to the collective, and r_{global} is the team score. The supply chain exhibits a *resource-competition* failure mode (hoarding stock starves downstream nodes), while MPE exhibits a *coordination-failure* mode (herding prevents coverage)—structurally distinct routes to the same collapse boundary. The collapse boundary is recovered in both environments we measure—the supply chain (PPO) and MPE (MAPPO). To make MPE a fair test of the behavioural decomposition rather than of the algorithm's capacity to coordinate, we set the herding scale so the null boundary lands in the interior (w_{crit}=0.74, rather than pinned against the centralization ceiling) and ran the full loss-aversion sweep $\lambda \in \{1,\ldots,7\}$ at α=β=0.88 (1,760 runs across the decomposition and λ-fill sweeps, 20 seeds per cell; Figure 6; cross-setting summary in Figure 5).

The *loss-aversion* effect replicates. On MPE (full λ=1…7 sweep, 20 seeds per cell), w_{crit} rises with loss aversion from 0.78 (λ=1) to 0.89 (λ=7)—rising in envelope but non-monotone at intermediate λ (the λ=2, 4 points dip to within seed noise of the null), so the effect is read as a direction, not a strict ordering. The loss-aversion excess over the null is $+0.158$ $[+0.108, +0.212]$ at λ=7 and $+0.101$ $[+0.055, +0.150]$ at λ=3, both significant and the latter quantitatively matching the supply-chain $+0.089$. The qualitative law—loss aversion raises the centralization required to avert collapse—thus holds across structurally distinct environments and training algorithms (a third environment, Level-Based Foraging, is added in Section ; Figure 5).

The *curvature* effect does *not* replicate. On MPE the curvature excess is $+0.042$ $[-0.009, +0.092]$, statistically indistinguishable from the null and opposite in sign to the supply-chain -0.100; probability weighting appears to facilitate coordination only in the supply-chain substrate. The exact *functional form* of the loss-aversion law is likewise environment-specific: the affine $w_{\text{crit}}(\lambda)$ that tracks the supply chain to RMSE 0.014 fits MPE only to RMSE 0.05, the MPE curve being steeper at high λ as the boundary approaches the centralization ceiling. We therefore claim cross-environment generality for the collapse boundary and for the *direction* of the loss-aversion effect—the paper's central behavioural claim—while reporting the curvature sign and the precise functional form as supply-chain-specific.

A third environment: Level-Based Foraging

MPE shares MAPPO's shared-global reward; to probe a third, structurally distinct domain we add Level-Based Foraging (LBF), a standard cooperative-MARL benchmark (Christianos et al., 2020) with a *public-goods* incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride and the team coordinates only as the global reward comes to dominate. LBF is natively discrete and unrelated to logistics or navigation; we train it under centralized-critic MAPPO—matching MPE, so that across MPE and LBF the *algorithm* is held fixed and only

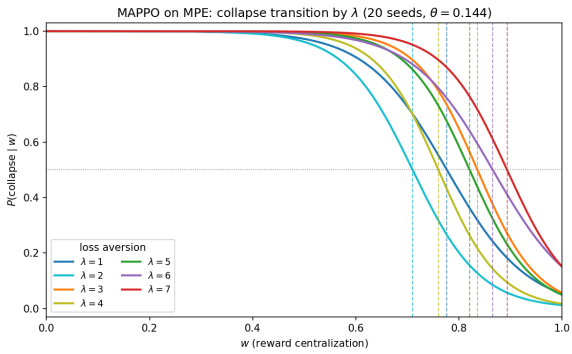


Figure 6: MAPPO on MPE Cooperative Navigation: fitted collapse-probability curves $P(\text{collapse} \mid w)$ for the full loss-aversion sweep λ=1, . . . , 7 (α=β=0.88, centralized critic, 50-step horizon, 20 seeds per cell; collapse = per-step landmark occupancy below θ=0.144, recalibrated from this run's null w=0/w=1 endpoints). The family shifts rightward as λ grows, with vertical dashed w_{crit} lines ordered accordingly—the same loss-aversion signature found in the supply chain, recovered under a different environment and a different (centralized-critic) algorithm.

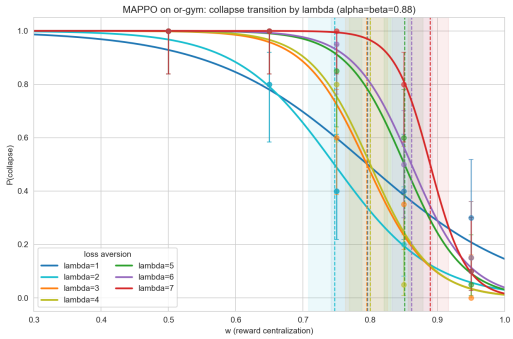


Figure 7: Centralized-critic MAPPO on the supply chain: fitted collapse-probability curves $P(\text{collapse} \mid w)$ for λ = 1, . . . , 7 (α=β=0.88, 300k steps, 20 seeds per cell, 700 runs). As under PPO, the family shifts rightward as loss aversion grows—each increment of λ raises the centralization required to avert collapse—confirming the effect is not an artifact of independent PPO. The PPO-vs-MAPPO $w_{\text{crit}}(\lambda)$ comparison is shown alongside MPE in Figure 5.

the *environment* varies—with the identical reward pipeline and full λ-sweep, and a collapse threshold set by the same collapsed-vs-coordinated midpoint used for MPE (1,600 runs, $w \in \{0, \ldots, 0.95\}$ including the w=0.90 resolver).

LBF yields the most complete decomposition outside the supply chain (Figure 5). The risk-neutral null has a low, tightly estimated boundary ($w_{\text{crit}} \approx 0.08$); probability-weighting curvature moves it *right* by $+0.18$ (impeding coordination)—*opposite* in sign to the supply chain's -0.10, reinforcing that the curvature sign is substrate-specific. Loss aversion then drives a large rightward shift: w_{crit} rises from 0.26 at λ=1 to 0.83 at λ=4, monotone with non-overlapping low- versus high-λ confidence intervals (the pre-registered criterion), and the direction is robust to the collapse threshold. For $\lambda \geq 5$ the boundary saturates against the centralization ceiling (the logistic w_{crit} extrapolates to $w \geq 1$), and we report those points as exploratory. To sharpen resolution in exactly the interval the high-λ boundaries occupy, we added an intermediate centralization level w=0.90 (a dedicated extension sweep) between the original 0.85 and 0.95; this tightens the mid-λ estimates and confirms the saturation, but $\lambda \geq 5$ still extrapolate to $w \geq 1$, so we continue to label those three points exploratory locations of a saturating boundary rather than precisely estimated values. As in MPE, the *direction* of the loss-aversion effect—the paper's central behavioural claim—replicates, while curvature sign and functional form remain substrate-specific. We make the latter quantitative: the supply chain's one-parameter affine law does *not* transfer to LBF. With g/c fixed from the λ=1 boundary, the best-fit slope ρ saturates at its bound and the law misfits LBF by RMSE 0.15 (versus 0.014 on the supply chain), because the LBF boundary rises more steeply in λ (already 0.85 by λ=4). This converts the substrate-specificity of the functional form from an assertion into a measured result—the *method* (a CI'd boundary that loss aversion shifts) generalizes, while the affine *constants and form* are substrate-specific.

Across algorithms: MAPPO on the supply chain

The MPE result varies the environment *and* the algorithm at once. To isolate the algorithm, we re-run the full loss-aversion sweep on the *identical* supply-chain substrate, grid, seeds, collapse label, and hyperparameters, changing only the learner from independent PPO to centralized-critic MAPPO (Yu et al., 2022) (700 runs; $\lambda \in \{1, \ldots, 7\}$, α=β=0.88, 5 values of w, 20 seeds). The actor consumes only its local observation; the critic consumes the joint state (CTDE). This isolates the effect of credit assignment from the behavioural incentive.

The loss-aversion law replicates under MAPPO. The critical centralization rises with loss aversion from w_{crit}=0.796 [0.742, 0.860] at λ=1 to w_{crit}=0.889 [0.856, 0.917] at λ=7, monotone from λ=2 onward (0.747, 0.795, 0.800, 0.851, 0.861, 0.889) and with non-overlapping λ=1 vs λ=7 confidence intervals (Figure 7; cross-setting summary in Figure 5). As in MPE, the *direction* generalizes while the absolute level does not: the MAPPO boundary sits above the PPO boundary at low λ (most visibly the λ=1 point, 0.796 vs 0.586), where MAPPO's transition is shallow and the w_{crit} estimate correspondingly noisier; the two boundaries converge by $\lambda \geq 4$ ($|\Delta w_{\text{crit}}| \leq 0.04$). We therefore report the loss-aversion direction and monotonicity as algorithm-independent on this substrate, and the absolute w_{crit} (especially at low λ) as algorithm-dependent.

Boundary Conditions: Two Ablations

The mean-field account (Section) makes the boundary contingent on two ingredients: a *continuous* policy gradient (the signal-to-noise crossing is a property of gradient flow), and a learner able to separate behavioural collapse from credit

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes (Section)
QMIX (centralised value)	discrete $K{=}11$	no
IQL (independent value)	discrete $K{=}11$	no
PPO (control)	discrete $K{=}11$	**no**

Table 6: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretisation of the continuous control—not the value-based paradigm—as what erases the boundary.

assignment (so that what w moves is the incentive, not the optimiser's capacity to coordinate). We ablate each. Both confirm the mechanism and delimit *where* the phenomenon lives.

Action representation

The mechanism predicts that destroying the continuous policy gradient should remove the boundary. Value-based learners require a discrete action space, so we discretised the supply-chain base-stock control into $K{=}11$ bins and ran an independent value learner (IQL) and a centralised monotonic-mixing learner (QMIX) on the identical reward pipeline, w-grid, and collapse label. Both produced a *flat, w-independent* collapse rate with no fittable w_{crit} in any condition. To isolate the cause—value-based learning vs. the discrete action space—we add a single-variable control: PPO, the learner that yields the boundary under continuous control, trained on the *same* $K{=}11$ discrete shell with everything else fixed. It too goes flat (Table 6). Holding the learner at PPO, discretisation *alone* erases the boundary, so the value-based nulls require no appeal to the learning paradigm.

We scope this precisely. The supply-chain action is a *continuous* base-stock level; coarse bins snap the policy to a stable interior level and never visit the near-threshold (s, S) region where collapse occurs, exactly as the continuous-gradient mechanism predicts. This is *not* a claim that discreteness per se removes the boundary: MPE (Section) has a *natively* discrete action space and retains it. The effect is therefore a coarse-discretisation boundary condition, substrate-specific in the same way the curvature sign is—and a confirmed prediction of the mechanism, not a negative result.

Credit assignment

The second ingredient is a learner that can coordinate at all on a shared-reward task, so that w_{crit} reflects the behavioural incentive rather than a credit-assignment failure. We ablate the centralised critic. On MPE Cooperative Navigation, an independent learner (PPO) never reaches the coordinated regime at *any* w: the collapse fraction is 1.0 across the entire w-range, including maximal centralisation (Figure 8), so behavioural collapse cannot be distinguished from the inability to assign credit. A centralised critic (MAPPO) on the identical environment recovers coordination—the collapse fraction falls to 0.35 and landmark occupancy crosses the calibrated

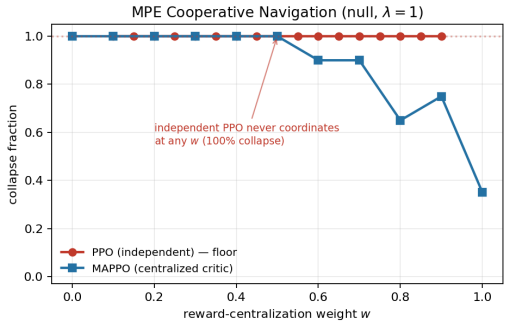


Figure 8: Credit-assignment ablation on MPE (null, $\lambda{=}1$). Independent PPO collapses at every w (a coordination floor: it cannot learn the task regardless of incentive), while the centralised-critic learner (MAPPO) crosses the boundary. The behavioural boundary is exposed only once credit assignment is handled.

threshold near $w{\approx}0.6$. We observe the same independent-learner floor on Level-Based Foraging, where PPO fails to coordinate while MAPPO reaches the coordinated regime. The boundary is therefore *measurable* only with a centralised critic on shared-reward coordination tasks; this is why we match a centralised critic to such environments (Section)—a necessity, not a convenience.

A Minority Sets the Boundary

The decomposition and its generalization treat the population as homogeneous—every agent carries the same λ. A deployed cooperative system rarely is: some participants are loss-averse, others are not. We therefore ask whether the boundary is an averaged, population-wide property, or whether a *minority* can set it—and whether *network position* matters. We assign $\lambda{=}7$ to selected nodes and $\lambda{=}1$ to the rest, holding the substrate, PPO configuration, collapse label, and grid identical to the main sweep ($\alpha{=}\beta{=}0.88$, 300k, 20 seeds, $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$).

The effect is large and strongly position-dependent (Figure 9, Table 7). A single *downstream* loss-averse node drives $w_{\mathrm{crit}} = 0.853$ $[0.849, 0.858]$—statistically indistinguishable from the *fully* loss-averse population ($w_{\mathrm{crit}} = 0.851$ $[0.847, 0.856]$): one biased node downstream sets the centralization requirement of the entire network. The same node placed *upstream* shifts the boundary far less, to $w_{\mathrm{crit}} = 0.726$ $[0.689, 0.763]$. Loss aversion therefore cascades *upstream*—a downstream node's bias propagates back along the ordering chain and dominates the system boundary—but not downstream, a directional asymmetry that mirrors the bullwhip structure of the substrate.

That a behavioral minority can move a system-level tipping point is itself established in evolutionary game theory via replicator selection (Mengesha et al., 2025); our contribution is the deep-MARL instantiation, where the effect emerges from gradient-learning dynamics and carries a concrete network-position signature absent from the EGT account. The operational reading is sharper than a population

average would suggest: in a mixed human–agent supply chain, the centralization the system needs is set by its most downstream loss-averse participant, not by the mean bias.

Table 7: Heterogeneous-population sweep (α=β=0.88; loss-averse nodes at λ=7, others λ=1; 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs. Homogeneous references: all risk-neutral ≈ 0.586, all loss-averse 0.851.

Loss-averse nodes	w_{crit} (95% CI)
one upstream	0.726 [0.689, 0.763]
one downstream	0.853 [0.849, 0.858]
two downstream	0.851 [0.847, 0.856]
all five (homogeneous)	0.851 [0.847, 0.856]

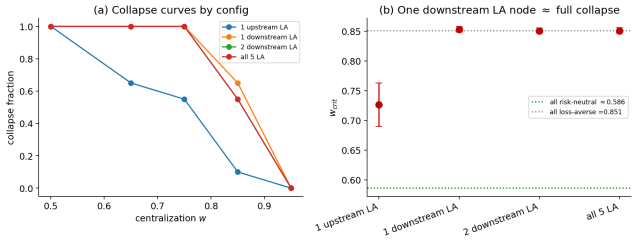


Figure 9: Heterogeneous cognitive profiles. (a) Collapse curves: a single downstream loss-averse node (orange) tracks the fully loss-averse population (red), while a single upstream node (blue) is far milder. (b) w_{crit} by configuration: one downstream loss-averse node already reaches the homogeneous loss-averse boundary; loss aversion cascades upstream but not downstream.

Discussion

That probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing its components (Lalmohammed et al., 2025; Ghaemi et al., 2024). Our null-model design isolates them: on the supply chain the beneficial curvature effect (-0.100, substrate-specific) is masked when combined with loss aversion, whose centralization-raising effect dominates and is the component we establish as general. The operational reading is concrete: a system with λ=7 agents—bracketing human loss-aversion estimates (Tversky and Kahneman, 1992)—requires $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents, so the additional 0.166 (Table 4) is a quantified cost of deploying loss-averse agents.

Two caveats sharpen this reading. On calibration, the canonical laboratory estimate of loss aversion is $\lambda \approx 2.25$ (Tversky and Kahneman, 1992); our sweep deliberately brackets the empirical human range, with λ=7 an upper, high-stakes bound rather than a typical value. The cost scales with λ (Table 4), so a domain whose participants sit nearer $\lambda \approx$ 2–3 incurs a smaller but still positive margin (the λ=3 excess is $+0.089$, against $+0.166$ at λ=7). On deployment, centralization is not free: privacy, communication cost, and organisational policy can cap the attainable w. Where the required w exceeds that cap, the boundary reframes the design problem—instead of only raising centralization, one can lower the agents' effective loss aversion (for instance through the training objective), treating λ as the second, behavioural lever the decomposition exposes.

The boundary is also not an averaged property. Section shows it is set by a *minority*: a single downstream loss-averse node moves the system boundary as far as a fully loss-averse population, cascading upstream along the ordering chain. The operational consequence is sharp—in a mixed human–agent supply chain, the required centralization is governed by the most downstream loss-averse participant, not by the population mean.

The sliding w_{crit} is a positive finding, not a pathology: the lazy-agent attractor is *metastable*, so agents caught below the isocline at finite T can in principle coordinate with more training—consistent with Weng and Lee (2026), where exploration breaks cooperation in shared-policy MARL before representations converge. The practical implication is that the minimum w depends on the training budget: tighter horizons demand higher centralization.

Yamaguchi (2025) construct an analogous (L, ρ) phase diagram for IQL; our results complement theirs by showing the phase structure depends not only on environmental complexity but on reward structure (w) and behavioral parameters (λ). More broadly, competing-bias tipping points are an established analytic phenomenon in multiagent dynamics: Capannoli et al. (2026) prove a metastable partial-adoption boundary in network opinion dynamics under competing stubbornness and disruptive biases, with an explicit critical threshold. Our contribution is the deep-MARL *learning-dynamics* analogue—a boundary set by the policy-gradient signal-to-noise ratio rather than a fixed update rule—which situates our phase-diagram framing within a recognised line of multiagent phase-transition results.

Several limitations should be noted. *Substrate scope.* The full quantitative characterization—the decomposition, the affine λ-law, metastability, and the heterogeneous cascade—uses one substrate (NetworkManagement-v1), chosen because its continuous action and per-agent economic reward make w_{crit} finely identifiable. We replicate the boundary and the loss-aversion *direction* in two further domains (MPE, LBF) and, to isolate algorithm from environment, re-run the full sweep under centralised-critic MAPPO on the same substrate (Section); MPE requires MAPPO because its shared-global reward defeats independent-PPO credit assignment, so we match the algorithm to each environment's learnability. The curvature effect is the one result that does *not* transfer—it lowers w_{crit} on the supply chain but is null/opposite on MPE—which the mechanism attributes to substrate-specific variance compression (Section); we therefore claim only the loss-aversion direction as cross-substrate. *Sweep scope.* The high-λ transitions (λ=5, 7) are steep and resolved by a single interior grid point—handled with Firth penalized fits (Table 3), though a finer grid above w=0.85 would tighten their intervals; we fix the demand distribution and lead times; the metastability result is finite-horizon (no asymptotic w_{crit}

bound estimated); and the heterogeneity sweep uses binary $\lambda \in \{1, 7\}$ rather than a continuous distribution (which the downstream cascade predicts is governed by its tail, not its mean). *Method scope conditions.* The approach requires an environment learnable without experience sharing and admitting a collapse label separable from mere task-unlearnability, excluding sparse, long-credit-horizon tasks. *Cascade is single-substrate—by structural necessity.* The downstream-dominated cascade (Section) is well-posed only where three properties co-occur: (i) a directed resource-flow ordering (defining "upstream"/"downstream"), (ii) genuine local-vs-global incentive tension (so w has a boundary to move), and (iii) a learning-reachable boundary. A survey of candidate second substrates found this conjunction rare—peer-symmetric tasks lack (i), regulation-style network-control tasks we examined lack (ii) and show no w-dependent boundary, and warehouse-delivery tasks violate (iii). We thus frame cross-substrate cascade not as an open conjecture but as a *falsifiable, structurally specified* prediction: in any domain satisfying (i)–(iii)—e.g. congested traffic networks or staged allocation with genuine hoarding—a minority of downstream loss-averse agents should set the system boundary. That most benchmarks violate at least one condition is itself a finding about where the phenomenon can live.

Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. Our central and most strongly supported finding is that *loss aversion* impedes coordination—raising w_{crit} ($+0.089$ at λ=3, within the human range; up to $+0.166$ at λ=7)—a direction we predict mechanistically a priori and confirm across three environments and two algorithms. Scoped to the supply-chain substrate, two further results sharpen the picture: a three-way decomposition shows probability-weighting curvature acting in the *opposite* direction (easing coordination by 0.100), and the boundary is set by a downstream loss-averse *minority* rather than the mean bias. We present the latter as a striking single-substrate result and frame it as a falsifiable prediction for any directed-flow domain, rather than a claim shown to generalize. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across further algorithms (QMIX) and environments (SMAC), upgrading the mean-field account (Section) from a calibrated mechanism to a quantitative prediction of w_{crit} via direct measurement of the policy-gradient signal-to-noise ratio (Section), adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous-λ populations.

Notation and Abbreviations

Symbol	Meaning
w	reward centralization weight, $\in [0, 1]$
w_{crit}	critical w where $P(\text{collapse})$=0.5
λ	loss-aversion coefficient (λ>1)
α, β	CPT value-function curvature (gain/loss)
ρ	loss-domain share of marginal incentive, $\in [0, 1]$
g, c	marginal global benefit / local cost of cooperation
$D(w), \Theta(w)$	Langevin drift / diffusion of the policy mean
$\text{SNR}(w)$	$D(w)/\sqrt{\Theta(w)}$, escape signal-to-noise ratio
σ, η, B	exploration scale / learning rate / batch size
$\Delta U, \tau$	escape barrier / Kramers first-passage time
T	training horizon (environment steps)
$\bar{S}, \bar{\pi}$	mean base-stock level / mean profit (collapse label)

Abbreviations. MARL (multi-agent reinforcement learning); CTDE (centralized training, decentralized execution); CPT (cumulative prospect theory); PPO (proximal policy optimization); MAPPO (multi-agent PPO with a centralized critic); MPE (multi-agent particle environments); SNR (signal-to-noise ratio); MLE (maximum-likelihood estimate); CI (confidence interval).

Claim–Evidence Ledger

Table 8 states exactly how broadly each headline claim is supported, so the scope of every claim is legible at a glance. The loss-aversion direction is the generalizing core (three environments, two algorithms); the curvature sign and the downstream-minority cascade are supply-chain-specific, the latter framed as a falsifiable prediction for directed-flow domains (Section).

Table 8: Each headline claim and the breadth of evidence behind it. Environments: SC (supply chain), MPE, LBF. Algorithms: PPO, MAPPO.

Claim	Envs	Algos
Sharp collapse boundary w_{crit} exists	SC, MPE, LBF	PPO, MAPPO
Metastability (w_{crit} slides with T)	SC	PPO
Loss aversion *raises* w_{crit} (dir.)	SC, MPE, LBF	PPO, MAPPO
Affine λ-law (functional form)	SC	PPO
Mechanism fixes sign a priori	SC, entropy abl.	PPO
Curvature *lowers* w_{crit} (sign)	SC (opp. on LBF)	PPO
Downstream-minority cascade	SC only	PPO

Threshold Robustness

We verify that the collapse definition (Equation (3)) is robust to perturbation of its thresholds. We re-estimate w_{crit} on Exp A @300k data under nine variants of the definition, varying $\bar{S}$ threshold over $\{8, 10, 12\}$ and $\bar{\pi}$ threshold

over $\{-102.5, -127.1, -152.5\}$. Three variants with threshold $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1, so no run satisfies this condition); they are excluded. Across the remaining six estimable variants, the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the confidence interval width. We conclude that the collapse definition is robust to $\pm 20\%$ threshold perturbation.

Pre-registered Test of the Signal-to-Noise Mechanism

The mean-field account (Section) calibrates the magnitude of w_{crit} rather than predicting it, because the relevant drift and diffusion coefficients are properties of the stochastic learning dynamics, not static quantities of the cost matrix. We pre-register here a direct measurement that would upgrade the magnitude from calibrated to predicted, and we fix its analysis before running it.

Observable. The mean-field model treats the policy's mean base-stock action μ as a Langevin variable, $\mathrm{d}\mu = D(w)\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W$. During PPO training we log μ once per update and, over a fixed early-training window, estimate the drift $D(w)$ (mean increment) and diffusion $\Theta(w)$ (increment variance / 2), forming $\mathrm{SNR}(w) = D(w)/\sqrt{2\Theta(w)}$.

Prediction and decision rule. We sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition (α=β=1, λ=1), 20 seeds per w, 150k steps, and compare against the *matched-horizon* boundary $w_{\text{crit}} = 0.822$ (Section). The mechanism is corroborated iff the measured signal-to-noise crossing's bootstrap 95% CI contains 0.822.

Outcome. The result is mixed, and we report it in full. The *collapse-fraction* of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly—a positive robustness check on the boundary itself. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localise w_{crit}. The reason is basin geometry: PPO initialises above the cooperative optimum ($S{\approx}250$ vs $S{\approx}150$), and collapses often plunge through the contested band too quickly to be sampled, biasing the conditioned drift toward survivors. We therefore do *not* claim a magnitude-level prediction; the boundary magnitude remains calibrated, as stated in Table 5. The mechanism section stands on its sign-level and functional-form predictions, which are unaffected. The driver and analysis (with the decision rule encoded) are released with the code so the test can be re-run with alternative observables (e.g. a longer horizon matched to $w_{\text{crit}} = 0.666$, or a basin-flow statistic).

Exploration-Noise Ablation

A further pre-registered experiment probes the boundary along an axis the main analysis holds fixed: the exploration noise of the learner. It reuses the identical substrate, PPO configuration, and collapse label, and provides the direct empirical support for the signal-to-noise mechanism (Section). (The heterogeneous-population result is promoted to the main text, Section .)

Exploration noise (entropy coefficient). The mean-field account (Section) predicts that w_{crit} is set by the signal-to-noise ratio of the policy gradient, so increasing the exploration term should raise it. We sweep the PPO entropy coefficient at the structural condition (α=β=1, λ=1, 300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$). The boundary rises monotonically with exploration noise (Figure 10, Table 9): $w_{\text{crit}} = 0.685$ at ent=0 (reproducing the null baseline exactly), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at *every* tested w (w_{crit} extrapolates above 1). This is direct empirical support for the signal-to-noise mechanism: the location of the boundary is not a function of the behavioral bias alone but also of the exploration noise that competes with the diluted cooperative signal, exactly as (9) predicts.

Table 9: Entropy-coefficient ablation (α=β=1, λ=1, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

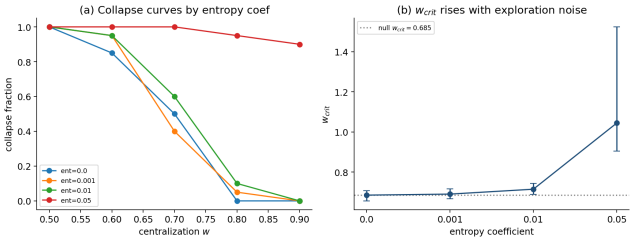


Figure 10: Exploration-noise ablation. (a) Collapse curves shift right as the entropy coefficient grows. (b) w_{crit} rises monotonically with exploration noise, corroborating the signal-to-noise mechanism (Section); ent=0 reproduces the null boundary 0.685.

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